

SCI225 Week 1 — Complete Study Guide

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Chapters 2, 3, 4 · Genes & Genetic Diseases · Epigenetics · Altered Cellular & Tissue Biology · Fluids, Electrolytes & Acid–Base

Study Spine — for every concept ask:

1. What is it? 2. What causes it? 3. How does it present? 4. What's the common student trap?

MECHANISM

WHY

COMPENSATION

PATHOLOGY

MISSTEP

ACRONYM

EXAM TIP

Course alignment

This guide maps directly to the three learning objectives that anchor the Week 1 exam:

- **LO2** — Understand the vast array of human genes and genetic diseases (Units 1)
- **LO3** — Define epigenetic and related genetic diseases (Unit 2)
- **LO4** — Define altered states of both cellular and tissue biology (Units 3–5, including fluids, electrolytes, and acid–base)

Every practice question in this guide was written for teaching. They mirror the *style* and *depth* of NCLEX-format items (single-best answer, select-all-that-apply, ordering, matching, drop-down) but are not copies of exam questions. Use them to test whether you can apply the concept, not memorize it.

Unit 1 • Genes & Genetic Diseases Ch 2 • LO2

1.1 Chromosomal Disorders — Numbers vs Structure

Human cells normally have **46 chromosomes (23 pairs)**: 22 autosomes + 1 sex pair. Abnormalities are either in **number** (aneuploidy) or in **structure** (deletion, duplication, translocation, inversion).

MECHANISM Nondisjunction is the single most important genetic concept for this exam. It is the *failure of homologous chromosomes (meiosis I) or sister chromatids (meiosis II) to separate*. The resulting gamete carries either an extra chromosome (n+1) or is missing one (n–1). After fertilization, the embryo is trisomic (3 copies) or monosomic (1 copy).

WHY Why maternal age matters so much. Oocytes start meiosis *before birth* and arrest at prophase I. They can sit paused for 30–45 years before being ovulated. The longer the arrest, the more the meiotic spindle degrades. Older spindles = more nondisjunction = higher trisomy risk. This is why maternal age is the #1 risk factor for trisomy 21, not paternal age.

The four chromosomal disorders you must know

Disorder	Karyotype	Key features	Classic exam clue
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Down syndrome (Trisomy 21)	47,XX +21 or 47,XY +21	Intellectual disability, flat facial profile, epicanthic folds, protruding tongue, low-set ears, single palmar (simian) crease, congenital heart defects (AV canal)	"Child with developmental delay, epicanthic folds, protruding tongue"
Turner syndrome (Monosomy X)	45,X	Female, short stature, webbed neck, coarctation of aorta, streak ovaries → infertility, no/delayed puberty	"14-year-old girl with short stature and delayed puberty"
Klinefelter syndrome	47,XXY	Tall male, small firm testes, gynecomastia, infertility, learning difficulties, delayed puberty	"14-year-old adolescent evaluated for delayed puberty, tall, gynecomastia"
Patau (Trisomy 13) / Edwards (Trisomy 18)	47 +13 / 47 +18	Severe defects, most die in infancy; cleft lip/palate (13), rocker-bottom feet (18)	"Newborn with multiple anomalies, short survival"

ACRONYM / MNEMONIC The Trisomy Rule of Sex + Age:

- Autosomal trisomy (13, 18, 21) → severe; 21 is most survivable
- Sex chromosome aneuploidy (XXY, X, XYY, XXX) → milder; often diagnosed at puberty
- **Maternal age** ↑ → nondisjunction ↑ → trisomy risk ↑

⚠ COMMON MISSTEP Three traps students fall into:

1. Confusing **deletion** (missing piece) with **trisomy** (extra chromosome). Down syndrome is an *extra* chromosome, not a deletion.
2. Confusing **single-gene mutation** with **chromosomal trisomy**. Trisomy 21 is not a mutation in one gene — it's a whole extra chromosome.
3. Thinking Turner is "a girl with extra X" — no, Turner is **45,X** (missing one). Klinefelter is the "extra X in a male" (47,XXY).

1.2 Single-Gene Disorders — Inheritance Patterns

Autosomal Recessive (AR)

- Both parents are typically **unaffected carriers**
- 25% affected · 50% carrier · 25% unaffected
- Affects **both sexes equally**
- Often "skips" generations
- **Examples:** Cystic fibrosis (CFTR), sickle cell, Tay-Sachs, PKU, albinism

X-Linked Recessive

- Gene on X chromosome
- **Males express it fully** (only one X — hemizygous)

Autosomal Dominant (AD)

- Only **one affected allele needed**
- Each child of affected parent has 50% chance
- Affected in **every generation**
- **Examples:** Huntington, Marfan, familial hypercholesterolemia, neurofibromatosis, polycystic kidney disease (adult)

Key reasoning pattern

- Mostly males affected, mom is healthy carrier → **X-linked recessive**
- Disease skips a generation, affects both sexes → **autosomal recessive**

- Females are usually **carriers** (rarely affected unless both X's are mutated)
- **No male-to-male transmission** — fathers pass Y, not X, to sons
- **Examples:** Duchenne muscular dystrophy, hemophilia A & B, color blindness, G6PD deficiency
- Every generation affected → **autosomal dominant**
- "Mom's brother died of the same disease" is the classic X-linked giveaway

High-yield single-gene diseases

Disease	Pattern	Defective gene / product	Clinical hallmarks
Cystic fibrosis	AR	CFTR chloride channel	Thick mucus → recurrent lung infections, pancreatic insufficiency, salty sweat, infertility
Phenylketonuria (PKU)	AR	Phenylalanine hydroxylase	↑ phenylalanine → intellectual disability, musty odor; treat with Phe-restricted diet
Sickle cell	AR	Hemoglobin S (β-globin)	Sickling under hypoxia → vaso-occlusion, hemolytic anemia, pain crises
Tay-Sachs / Lysosomal storage	AR	Hexosaminidase A (and many others)	Enzyme deficiency → material builds up inside lysosomes → cells die
Huntington disease	AD	HTT gene (CAG repeats)	Mid-adult onset: chorea, psychiatric symptoms, dementia; anticipation in later generations
Duchenne MD	X-linked R	Dystrophin protein	Boys only, progressive proximal weakness (Gower sign), calf pseudohypertrophy
Hemophilia A	X-linked R	Factor VIII	Boys, bleeding into joints, prolonged aPTT

MECHANISM Cystic fibrosis — follow the chain. The exam loves this sequence:

CFTR gene mutation → defective chloride channel in epithelium → abnormal Cl⁻/water transport → thick, dehydrated mucus → impaired mucociliary clearance → recurrent lung infection

Know this order. Ordering questions love this exact cascade.

★ **EXAM TIP** If you see "mom's brother died of the same disease" → think **X-linked recessive**. The disease came through the unaffected (carrier) mother. If you see "both parents healthy carriers" → **autosomal recessive**. If you see "every generation affected, adult onset" → **autosomal dominant**.

1.3 Multifactorial Inheritance

Many common diseases are caused by **multiple genes + environment**, not a single mutation. These cluster in families but do not follow Mendelian ratios.

MECHANISM Multifactorial conditions have a *threshold of liability*: you need both enough genetic predisposition AND enough environmental trigger. Genes alone aren't destiny; lifestyle alone isn't destiny either.

Common multifactorial conditions: hypertension, type 2 diabetes, coronary artery disease, cleft lip/palate, neural tube defects, schizophrenia, most congenital heart defects.

⚠ **COMMON MISSTEP** If a vignette says "his father and two siblings also developed HTN as adults," the answer is **multifactorial inheritance** — NOT autosomal dominant, even though it clusters in the family. Multifactorial = genes + environment interacting.

Practice Questions — Genetic Diseases

Q1. A 35-year-old primigravida asks about her risk of having a child with Down syndrome compared with a 25-year-old. The best explanation is:

- A. A 35-year-old has a higher chance of producing sperm with chromosome errors
- B. Aging oocytes spent longer arrested in meiosis, and their spindle apparatus is more error-prone
- C. Older mothers have more environmental DNA damage during pregnancy
- D. Down syndrome is autosomal recessive, and older parents are more likely to be carriers

Show answer

B. Oocytes begin meiosis in fetal life and arrest at prophase I until ovulation. Decades of arrest degrade the meiotic spindle, raising nondisjunction risk. A (sperm) is wrong — maternal, not paternal, age is the driver. C (environment) is not the primary mechanism. D is wrong — Down syndrome is a trisomy, not recessive inheritance.

Q2. A 16-year-old girl presents with short stature, a webbed neck, and absent breast development. Karyotype is 45,X. Which pathophysiologic process produced her condition?

- A. A gene mutation on the X chromosome
- B. Failure of sex chromosomes to separate during meiosis, producing a gamete with no X
- C. Genomic imprinting silenced the maternal X
- D. A translocation moved part of the X onto an autosome

Show answer

B. Turner syndrome is monosomy X from nondisjunction. A is wrong (not a single-gene problem). C is irrelevant to Turner. D describes a different structural rearrangement. Always connect 45,X → missing one sex chromosome → nondisjunction.

Q3. A couple who are both unaffected cystic fibrosis carriers ask about the chance of having an affected child. The best answer is:

- A. 0% — both parents are healthy
- B. 25%
- C. 50%
- D. 100%

[Show answer](#)

B — 25%. In autosomal recessive disease with two carriers: 25% affected (both mutant alleles), 50% carriers (one mutant), 25% unaffected (neither). Memorize the 1:2:1 Punnett pattern for any AR cross.

Q4. Select all that apply. Which features are expected in Duchenne muscular dystrophy?

- A. X-linked recessive inheritance
- B. Affects predominantly males
- C. Carrier mothers are usually unaffected
- D. Affected fathers pass the condition to their sons
- E. Caused by a defective dystrophin protein

[Show answer](#)

A, B, C, E. D is wrong — fathers pass the Y to sons (no X-linked disease goes father → son). Carrier mothers usually have one normal X, so they're unaffected. Dystrophin loss → progressive muscle fiber breakdown.

Q5. Order. Place these events in sequence as they occur in a child with cystic fibrosis, starting with the root cause:

- Thick mucus blocks the airways
- Repeated lung infections develop
- Mutation in the CFTR gene
- Abnormal chloride and water transport at the epithelial surface

[Show answer](#)

Correct order:

1. Mutation in the CFTR gene
2. Abnormal chloride and water transport at the epithelial surface
3. Thick mucus blocks the airways
4. Repeated lung infections develop

Gene → protein → transport → mucus → infection. Always start at the DNA level for ordering questions.

Q6. Matching. Pair each condition with its karyotype.

Condition	Karyotype
Down syndrome	_____
Turner syndrome	_____
Klinefelter syndrome	_____
Trisomy 18 (Edwards)	_____

Choices: 45,X · 47,XX +18 · 47,XY +21 · 47,XXY

[Show answer](#)

Down → 47,XY +21 · Turner → 45,X · Klinefelter → 47,XXY · Edwards → 47,XX +18.

Trick: Turner is the only *monosomy* compatible with life; Klinefelter is "extra X in a male."

Q7. A 50-year-old man has severe hypertension. His father, brother, and sister were all diagnosed with HTN as adults. This pattern most likely reflects:

- A. Autosomal dominant inheritance
- B. X-linked recessive inheritance
- C. Multifactorial inheritance
- D. Chromosomal trisomy

[Show answer](#)

C. HTN clusters in families but doesn't follow Mendelian ratios — multiple genes + diet + stress + obesity all contribute. The trap is picking "autosomal dominant" because it runs in the family. Familial clustering alone ≠ AD.

Q8. Dropdown. A child has a deficiency of a lysosomal enzyme. Inside cells, [select: mucus / fluid / substrate material / DNA] accumulates, leading to [select: apoptosis / swelling and death of many cells / cell membrane repair / enhanced growth].

[Show answer](#)

Dropdown 1: **substrate material**. Dropdown 2: **swelling and death of many cells**.
Lysosomes normally break substrates down; missing the enzyme = the material that should have been digested just builds up inside lysosomes until cells rupture. Tay-Sachs is the classic example.

Unit 2 • Epigenetics Ch 3 • LO3

Epigenetics = heritable changes in gene *expression* that do **not** alter the DNA sequence. The code stays the same; which genes are turned on or off changes.

MECHANISM The three main epigenetic mechanisms:

1. **DNA methylation** — methyl groups attached to cytosines (especially at gene promoters) usually *silence* gene expression.
2. **Histone modification** — chemical tags on histones change how tightly DNA wraps around them, affecting gene accessibility.
3. **Non-coding RNA** — small RNAs (like miRNA) regulate which mRNAs get translated.

WHY Why epigenetics matters clinically: Environment (diet, toxins, stress, hormones) can change gene *expression* without changing the gene itself. These changes can be inherited across cell divisions, and some can even cross generations. Identical twins with the same DNA can develop different diseases because they accumulated different epigenetic marks.

Genomic Imprinting

MECHANISM Normal cells have two copies of most genes — one from mom, one from dad. **Imprinting** means only *one parental copy* is expressed; the other is silenced by methylation. If the active copy is lost or inactivated, there is no functional gene at all in that region — even though one copy "looks" intact.

Two textbook examples from opposite parents, same region (chr 15):

Disorder	Deleted from	Features
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Prader-Willi syndrome	Paternal chr 15 (maternal copy was already silenced by imprinting)	Hypotonia, hyperphagia → obesity, short stature, intellectual disability, hypogonadism
Angelman syndrome	Maternal chr 15 (paternal copy was already silenced)	"Happy puppet" — severe intellectual disability, ataxia, inappropriate laughter, seizures

★ **EXAM TIP** Think of imprinting like a light switch wired with only one working input. If the wire that actually carries current gets cut, the switch does nothing — even though the other wire still exists. That's why *which parent* contributed the deletion decides which disease you get.

Developmental Programming (the fetus and long-term epigenetics)

PATHOLOGY Maternal diabetes in pregnancy. Hyperglycemia crosses the placenta → fetal pancreas produces more insulin → excess growth, increased adipose tissue (macrosomia). Beyond birth, the fetus's *epigenome is reprogrammed*, increasing lifetime risk of obesity, metabolic syndrome, and type 2 diabetes. This is a clinically important case of epigenetic inheritance of disease risk without a DNA mutation.

WHY Environmental exposures during early pregnancy (air pollution, famine, toxins) can alter fetal DNA methylation without causing a mutation. The baby's gene expression is altered for life. This is why "did not cause a genetic mutation" does not mean "had no effect" — the exam will test this distinction.

Epigenetics in Cancer

PATHOLOGY Silencing of tumor suppressor genes. Many cancers *don't delete* tumor suppressor genes — they silence them by hypermethylation of the promoter. The gene is still present in the DNA, but it's not expressed. Without tumor suppressors, damaged cells escape apoptosis and proliferate.

⚠ **COMMON MISSTEP "Epigenetics = genetic mutation."** No — that's the most common trap. In epigenetics, the DNA sequence is unchanged; only gene *expression* is altered. On SATA questions, distractors that say "the baby's DNA was mutated" are almost always wrong when the stem specifies "epigenetic change."

ACRONYM / MNEMONIC "SAME DNA, DIFFERENT LIVES." Use this for any identical-twin question where one twin develops disease and the other doesn't. The answer is almost always **epigenetic modification** from different environmental exposures, not a new mutation.

Practice Questions — Epigenetics

Q9. Identical twins have the same DNA. One develops type 2 diabetes at age 45; the other does not. Which explanation best accounts for this difference?

- A. One twin's DNA mutated after birth
- B. Different environmental exposures produced different epigenetic marks that altered gene expression

- C. One twin inherited a new gene the other did not
- D. The twins are not truly identical

[Show answer](#)

B. Same DNA + different lives = epigenetic divergence (methylation, histone modification, miRNA) altering which genes are expressed. A is wrong because the DNA sequence does not change. C is impossible — identical twins start with identical DNA. D contradicts the stem.

Q10. A patient's lung cancer biopsy shows that a tumor suppressor gene is no longer expressed, but the gene itself is present and not mutated. Which mechanism best explains this?

- A. A translocation moved the gene to a different chromosome
- B. The promoter region became heavily methylated, silencing transcription
- C. The gene was deleted entirely
- D. Nondisjunction produced an extra chromosome

[Show answer](#)

B. Promoter methylation is a classic *epigenetic* mechanism of tumor suppressor silencing — the gene is intact, but expression is shut off. A and C involve structural DNA change. D would cause aneuploidy, not loss of one gene's expression.

Q11. A pregnant patient with poorly controlled type 2 diabetes is told that her infant may be at lifelong increased risk for obesity and diabetes. Which statement best explains this?

- A. High maternal glucose causes DNA mutations in the fetus
- B. Maternal hyperglycemia alters fetal gene expression through epigenetic programming
- C. Insulin crosses the placenta and directly mutates fetal DNA
- D. Maternal diabetes always causes Down syndrome

[Show answer](#)

B. This is *developmental programming*: intrauterine environment reshapes fetal DNA methylation and gene expression, raising lifelong metabolic disease risk. No mutation is required.

Q12. A child has Prader-Willi syndrome from a deletion of the paternal chromosome 15. Which principle best explains the disease?

- A. X-inactivation
- B. Nondisjunction
- C. Genomic imprinting — gene expression depends on parent of origin
- D. Translocation

[Show answer](#)

C. The maternal copy of the critical chr 15 region is already silenced by imprinting. If the paternal copy is deleted, no functional gene is left. Same deletion on the maternal side would cause Angelman syndrome instead.

Q13. A pregnant patient exposed to heavy air pollution asks how the exposure could affect her baby even if it does not cause a genetic mutation. The best answer is:

- A. "There is no effect unless the DNA is mutated."
- B. "Exposures can change which fetal genes are expressed, through epigenetic modifications like methylation, without changing the DNA sequence."
- C. "Only large radiation exposures matter in early pregnancy."
- D. "Air pollution only affects maternal health, not fetal development."

Show answer

B. The question tests whether you understand that *epigenetic* effects are real, heritable, and do not require DNA sequence changes.

Unit 3 • Cellular Adaptation & Injury Ch 4 • LO4

Cells respond to stress in predictable ways. If the stress is mild and gradual, cells **adapt**. If the stress is severe or sustained, cells become **injured** and may die. The exam heavily tests this progression.

3.1 The Four Cellular Adaptations

MECHANISM

HYPERTROPHY

Cells get *bigger* (same number).
Trigger: ↑ functional demand or hormonal stimulation.
Examples: cardiac muscle with HTN (LVH), skeletal muscle with weight training, pregnant uterus.

MECHANISM

HYPERPLASIA

Cells get *more numerous* (same size each).
Trigger: hormonal or compensatory.
Examples: breast tissue in pregnancy, endometrium (estrogen), liver regeneration, BPH.

MECHANISM ATROPHY

Cells get *smaller* (and/or fewer).
Triggers: disuse, denervation, loss of blood supply, loss of hormonal stimulation, aging.
Examples: casted limb, post-stroke paralysis, prolonged ventilation, postmenopausal endometrium.

MECHANISM METAPLASIA — one differentiated cell type is replaced by another, in response to chronic stress. Reversible but often maladaptive.

Classic example: chronic smoker — normal ciliated columnar epithelium in the bronchi is replaced by stratified squamous. Squamous cells are tougher but can't clear mucus. *Barrett esophagus* is another (squamous → columnar in response to chronic acid reflux).

PATHOLOGY DYSPLASIA — disordered, abnormal-looking cells. Not a normal adaptation. Often the step that precedes cancer (e.g., cervical dysplasia → cervical carcinoma in situ → invasive).

ACRONYM / MNEMONIC AHHAM the 5 responses:

Atrophy (smaller) · Hypertrophy (bigger cells) · Hyperplasia (more cells) · Adaptive metaplasia · Maladaptive dysplasia (toward cancer).
Or remember the direction: *size* (hypertrophy/atrophy), *number* (hyperplasia), *type* (metaplasia), *order* (dysplasia).

⚠ COMMON MISSTEP

- **"Hypertrophy and hyperplasia are the same."** No — size vs number. Cardiac and skeletal muscle cannot hyperplase (they are post-mitotic), so they can only hypertrophy.

- **"Atrophy is always pathologic."** No — thymic atrophy with age and uterine atrophy after menopause are physiologic.
- **"Metaplasia is cancer."** Not yet — it's adaptive and reversible. Dysplasia is the step that starts drifting toward malignancy.

3.2 Cellular Injury and Death

MECHANISM Hypoxia is the #1 cause of cell injury. ↓ O₂ delivery → mitochondria can't make ATP → membrane pumps fail → Na⁺ and water enter → cell swelling (reversible). If hypoxia continues → Ca²⁺ floods in → enzymes activate → membranes rupture → irreversible injury → necrosis.

Hypoxia → ↓ ATP → Na/K pump fails → cell swells → anaerobic metabolism → ↑ lactate, ↓ pH → mitochondrial Ca²⁺ influx → membrane rupture → necrosis

Necrosis vs Apoptosis — the high-yield comparison

Feature	Necrosis	Apoptosis
Trigger	Injury (ischemia, toxin, trauma)	Programmed — developmental, immune, DNA damage
Cell membrane	Ruptures, contents spill	Stays intact, forms apoptotic bodies
Inflammation	Yes (because contents leak)	No (neighbors phagocytose bodies quietly)
ATP requirement	No — ATP is depleted	Yes — ATP required (it's active!)
Examples	MI, stroke infarct, gangrene, pancreatitis	Embryonic web removal, menstrual shedding, CD8 killing a virus-infected cell

★ **EXAM TIP** Apoptosis is "clean" death. The cell dismantles itself neatly without inflaming the neighborhood. Necrosis is "messy" death with inflammation. This is why most textbook disease processes cause necrosis — the inflammation is part of the clinical picture.

WHY Lactate rises in **hypoxic cells** because the cell shifts from aerobic to anaerobic metabolism. High lactate on a lab report is a red flag for tissue hypoperfusion and impending irreversible injury — often seen in septic or cardiogenic shock.

Practice Questions — Adaptation & Injury

Q14. A patient with long-standing hypertension has thickening of the left ventricular wall on echocardiogram. This change represents:

- Hyperplasia
- Hypertrophy
- Metaplasia
- Atrophy

[Show answer](#)

B. Cardiac myocytes are post-mitotic — they cannot divide, so they respond to chronic pressure load by enlarging (hypertrophy). Hyperplasia (A) can't happen in cardiac muscle for this reason.

Q15. A chronic smoker's bronchial biopsy shows replacement of ciliated columnar epithelium with stratified squamous epithelium. This is:

- A. Hypertrophy
- B. Hyperplasia
- C. Metaplasia
- D. Dysplasia

[Show answer](#)

C. One differentiated cell type replaced by another = metaplasia. It's reversible if the smoker quits. If it progresses to disordered cells, that's dysplasia and increases cancer risk.

Q16. A 68-year-old mechanically ventilated for three weeks has lost significant skeletal muscle mass. This reflects which cellular adaptation?

- A. Atrophy from disuse and decreased nerve stimulation
- B. Hypertrophy from mechanical strain
- C. Metaplasia of muscle to connective tissue
- D. Apoptosis due to denervation

[Show answer](#)

A. Disuse + loss of neural input = atrophy. Skeletal muscle is particularly prone because movement is required to maintain mass. Stroke-related paralysis is another classic example.

Q17. Select all that apply. Which are features of apoptosis (NOT necrosis)?

- A. Requires ATP
- B. Cell membrane stays intact
- C. Causes surrounding inflammation
- D. Occurs normally during embryologic development
- E. Caused by ischemia

[Show answer](#)

A, B, D. Apoptosis is active (needs ATP), "quiet" (no inflammation), and physiologic (development, menstrual shedding). C and E describe necrosis.

Q18. A patient has severe shock with elevated serum lactate and metabolic acidosis. Which best explains the lactate elevation?

- A. Cells are producing extra lactate as fuel
- B. Hypoxia forces cells into anaerobic metabolism, which produces lactate as an end product
- C. The kidneys cannot secrete lactate
- D. Lactate is being absorbed from the GI tract

Show answer

B. Without O_2 , pyruvate is converted to lactate so glycolysis can continue. Rising lactate = tissue hypoperfusion and a warning that cells are near irreversible injury.

Q19. Order. Arrange the following hypoxic injury events in order:

- Mitochondrial Ca^{2+} influx and membrane rupture
- Na/K pump failure and cell swelling
- Oxygen delivery falls
- ATP production decreases
- Anaerobic metabolism produces lactate

Show answer

Correct order: O_2 delivery falls → ATP ↓ → Na/K pump fails + swelling → anaerobic metabolism + lactate → Ca^{2+} influx + membrane rupture. Order questions love this cascade — memorize it.

Unit 4 • Fluids & Electrolytes Ch 4 · LO4

4.1 Fluid Balance & Edema

MECHANISM Total body water $\approx$ 60% of body weight. $\frac{2}{3}$ is inside cells (ICF); $\frac{1}{3}$ is outside (ECF). ECF is split into interstitial fluid ($\frac{3}{4}$) and plasma ($\frac{1}{4}$). Water moves between compartments by osmosis — it follows solute.

Fluid movement at capillaries is governed by four forces (Starling):

- Capillary **hydrostatic** pressure — pushes water OUT of the vessel
- Capillary **oncotic** pressure (from albumin) — pulls water IN
- Interstitial hydrostatic + interstitial oncotic — opposite directions

WHY Four classic causes of edema — each is a Starling force gone wrong:

1. ↑ **hydrostatic pressure** — heart failure (congestion), venous obstruction
2. ↓ **oncotic pressure** — low albumin (liver failure, nephrotic syndrome, malnutrition)
3. ↑ **capillary permeability** — inflammation, burns, sepsis
4. **Lymphatic obstruction** — post-mastectomy, filariasis

PATHOLOGY **Liver failure + edema.** The liver makes albumin. Failing liver → ↓ serum albumin → ↓ capillary oncotic pressure → fluid leaks into interstitium → generalized edema and ascites. This is tested often.

Dehydration

Fluid loss (vomit, diarrhea, sweat, diuretic) → ↓ plasma volume → ↓ BP → baroreceptors + RAAS + ADH activate → vasoconstriction + Na/water retention → attempt to restore volume

4.2 Electrolyte Imbalances — The Big Three

Electrolyte	Normal	Critical role
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Sodium (Na ⁺)	135–145 mEq/L	Main ECF cation → controls osmolarity and water distribution → affects neuro
Potassium (K ⁺)	3.5–5.0 mEq/L	Main ICF cation → controls membrane excitability → affects heart
Calcium (Ca ²⁺)	8.5–10.5 mg/dL	Bone, muscle contraction, clotting → affects neuromuscular excitability

Sodium

	Hyponatremia (<135)	Hypernatremia (>145)
Causes	Excess water (water intoxication, SIADH), GI loss, diuretics	Water loss without electrolyte replacement, DI, burns
Symptoms	Nausea, headache, confusion, seizures → cerebral edema	Thirst, dry mucosa, lethargy, irritability, seizures

Potassium — always think the heart

	Hypokalemia (<3.5)	Hyperkalemia (>5.0)
Causes	GI loss (vomit, diarrhea), diuretics, insulin/β-agonists	Renal failure (CKD), K-sparing diuretics, acidosis, tissue breakdown
ECG	Flattened T waves, U waves, ST depression, PVCs	Peaked T waves , wide QRS, sine wave, V-fib/arrest
Neuromuscular	Weakness, cramps, paralysis, ↓ reflexes	Weakness, paresthesia, flaccid paralysis

★ **EXAM TIP** **Peaked T waves** → hyperkalemia. **U waves** → hypokalemia. These are the two ECG findings most likely on the exam. Both can cause fatal arrhythmias; hyperkalemia can arrest the heart in diastole.

PATHOLOGY **CKD patient with K 6.2 mEq/L.** CKD kidneys can't excrete potassium efficiently. Hyperkalemia is a leading cause of death in CKD. Expect peaked T waves on ECG.

Calcium — mirror image neuromuscular effects

	Hypocalcemia (<8.5)	Hypercalcemia (>10.5)
Causes	Hypoparathyroidism, renal failure (↓ vit D activation), pancreatitis	Hyperparathyroidism, malignancy (paraneoplastic), immobility
Symptoms	Tetany , Chvostek sign (facial twitch), Trousseau sign (carpopedal spasm), paresthesias	"Stones, bones, groans, psychiatric moans" — kidney stones, bone pain, abdominal pain, confusion

⚠ **COMMON MISSTEP** Three electrolyte traps:

- **Hyponatremia from overhydration ≠ dehydration.** Drinking too much water dilutes Na; the patient is overloaded, not dry.
- **Hyperkalemia's classic ECG sign is peaked T waves** — students often reverse this with U waves (which belong to hypokalemia).

- **Hypocalcemia causes tetany; hypercalcemia causes sedation** — NOT the other way around. Low Ca raises neuronal excitability.

Practice Questions — Fluids & Electrolytes

Q20. A patient with liver failure develops generalized edema. Laboratory studies show markedly reduced serum albumin. Which Starling force best explains the edema?

- A. Increased capillary hydrostatic pressure
- B. Decreased capillary oncotic pressure
- C. Increased tissue oncotic pressure
- D. Lymphatic obstruction

[Show answer](#)

B. Low albumin → low plasma oncotic pressure → water is not pulled back into the vessels → interstitium expands → edema/ascites.

Q21. A CKD patient has K^+ 6.3 mEq/L. Which ECG finding is most expected?

- A. U waves
- B. Prolonged PR with flat T waves
- C. Peaked T waves
- D. Q waves in leads II, III, aVF

[Show answer](#)

C. Peaked T waves are the classic early sign of hyperkalemia. A (U waves) belongs to hypokalemia. D (Q waves) is MI.

Q22. A patient with K^+ 2.8 mEq/L reports muscle weakness. The nurse notes PVCs on the monitor. The best explanation is:

- A. Hypokalemia lowers the resting membrane potential threshold, increasing cardiac ectopy
- B. Hypokalemia directly destroys the SA node
- C. Hypokalemia mimics hypercalcemia
- D. Hypokalemia increases ATP production in the heart

[Show answer](#)

A. Low K makes cardiac cells hyperpolarized and more excitable — PVCs and arrhythmias are typical. Severe hypokalemia can cause torsades.

Q23. A patient drinks large volumes of plain water after prolonged exertion and develops Na^+ 122 mEq/L with confusion. The pathophysiologic change is:

- A. Sodium loss from sweat alone
- B. Dilutional hyponatremia from excess free water with no electrolyte replacement
- C. Renal failure acutely
- D. Hyperaldosteronism

[Show answer](#)

B. Overhydration dilutes the ECF sodium. Low Na draws water into brain cells → cerebral edema → confusion or seizure. This is why sports drinks exist.

Q24. Matching. Match each electrolyte imbalance to its characteristic manifestation.

Imbalance	Manifestation
Hyperkalemia	_____
Hypokalemia	_____
Hypercalcemia	_____
Hypocalcemia	_____

Choices: Tetany, Chvostek sign · Peaked T waves · U waves, flat T · Kidney stones, confusion

[Show answer](#)

HyperK → peaked T waves · HypoK → U waves, flat T · HyperCa → kidney stones, confusion · HypoCa → tetany, Chvostek.

Q25. A 76-year-old man has 3 days of vomiting and diarrhea and now presents confused. Which electrolyte pattern is most expected?

- A. Low K, low Na, metabolic acidosis from diarrhea and metabolic alkalosis from vomiting (mixed)
- B. High Na and high K
- C. Low Ca only
- D. No electrolyte change

[Show answer](#)

A. Vomiting loses HCl (alkalosis, low K, low Cl) and diarrhea loses HCO₃ (acidosis, low K). Both deplete K and volume, so this patient classically has mixed electrolyte and acid-base abnormalities plus dehydration.

Unit 5 • Acid–Base Balance Ch 4 · LO4

5.1 ABG Fundamentals

Value	Normal	What it tells you
pH	7.35–7.45	Overall blood acidity. <7.35 = acidemia; >7.45 = alkalemia
PaCO ₂	35–45 mm Hg	Respiratory component. ↑ CO ₂ = respiratory acidosis; ↓ CO ₂ = respiratory alkalosis
HCO ₃ [−]	22–26 mEq/L	Metabolic component. ↓ HCO ₃ = metabolic acidosis; ↑ HCO ₃ = metabolic alkalosis
PaO ₂	80–100 mm Hg	Oxygenation (often tracks SaO ₂)

ACRONYM / MNEMONIC ROME — the quickest way to interpret ABGs:

Respiratory Opposite — pH and CO₂ move in *opposite* directions.

Metabolic Equal — pH and HCO₃ move in the *same* direction.

Example: pH ↓ 7.22, CO₂ ↑ 55 → both moving opposite → respiratory acidosis.

Example: pH ↓ 7.22, HCO₃ ↓ 12 → both moving equal → metabolic acidosis.

A 3-step method you can use on any ABG

1. **Look at the pH.** Is it acidemia or alkalemia?
2. **Match pH direction to CO₂ or HCO₃.** Whichever moves with the pH (in the same disease direction) is the primary problem.
3. **Check compensation.** Is the opposite system moving the right direction to pull pH toward normal?

5.2 The Four Acid–Base Disorders

Disorder	pH	Primary change	Compensation	Classic causes
Respiratory acidosis	↓	↑ PaCO ₂	Kidneys retain HCO ₃ (slow, days)	COPD, opioid overdose, neuromuscular weakness, hypoventilation
Respiratory alkalosis	↑	↓ PaCO ₂	Kidneys excrete HCO ₃	Anxiety, PE, high altitude, pain, early sepsis, mechanical over-ventilation
Metabolic acidosis	↓	↓ HCO ₃	Lungs blow off CO ₂ (Kussmaul breathing)	DKA, lactic acidosis (shock), diarrhea, renal failure, aspirin toxicity
Metabolic alkalosis	↑	↑ HCO ₃	Lungs retain CO ₂ (slow breathing)	Vomiting (loss of HCl), nasogastric suction, diuretics, antacid abuse

PATHOLOGY DKA — a classic metabolic acidosis. Insulin deficiency → fat breakdown → ketoacids → ↓ HCO₃ → ↓ pH. Compensation: deep, rapid **Kussmaul breathing** blows off CO₂ to raise pH. Expect glucose >300, HCO₃ <18, pH <7.30, positive ketones, fruity breath.

PATHOLOGY COPD exacerbation — a classic respiratory acidosis. Airway obstruction traps CO₂ → ↑ PaCO₂ → ↓ pH → confusion, somnolence. Chronic COPD patients compensate by retaining HCO₃ over days; acutely on chronic, pH drops further.

WHY Why does a person breathe fast and deep in DKA? The metabolic acidosis (low HCO₃) drives the body to blow off CO₂ through the lungs. Lower CO₂ pulls pH up toward normal. Kussmaul breathing is a *compensation*, not the cause.

⚠ **COMMON MISSTEP** Common ABG traps:

- **DKA patient breathing fast is NOT respiratory alkalosis.** It's metabolic acidosis with respiratory compensation. Always look at pH and HCO₃ first.
- **Vomiting causes metabolic alkalosis** (losing HCl), NOT acidosis. Diarrhea is the one that causes metabolic acidosis (losing HCO₃).
- **COPD = respiratory acidosis** (CO₂ retention). Students sometimes pick alkalosis because "the patient is sick" — read the PaCO₂.

Practice Questions — Acid–Base

Q26. A patient with DKA has pH 7.22, PaCO₂ 26, HCO₃ 10. The acid–base disturbance is:

- A. Respiratory acidosis with metabolic compensation
- B. Respiratory alkalosis with metabolic compensation
- C. Metabolic acidosis with respiratory compensation
- D. Metabolic alkalosis with respiratory compensation

[Show answer](#)

C. pH is low (acidosis). HCO_3 is low (matches pH, so metabolic — ROME). Low PaCO_2 is the lungs compensating by blowing off CO_2 (Kussmaul breathing).

Q27. A 65-year-old with COPD is admitted confused, shallow breathing, SaO_2 88%. ABG: pH 7.28, PaCO_2 62, HCO_3 30. The disturbance is:

- A. Respiratory acidosis
- B. Respiratory alkalosis
- C. Metabolic acidosis
- D. Metabolic alkalosis

[Show answer](#)

A. pH ↓ and PaCO_2 ↑ → respiratory acidosis (ROME: opposite directions). The mildly elevated HCO_3 is renal compensation. COPD patients retain CO_2 as airway obstruction worsens.

Q28. Order. Place these events in the correct order during respiratory failure in a patient who cannot eliminate enough CO_2 :

- pH decreases
- PaCO_2 rises
- Ventilation is inadequate
- Kidneys begin retaining HCO_3 over days

[Show answer](#)

Correct order: Ventilation inadequate → PaCO_2 rises → pH decreases → kidneys retain HCO_3 (renal compensation over days). The respiratory failure causes the acidosis; renal compensation is slow.

Q29. A patient with prolonged vomiting presents with pH 7.49, HCO_3 34, PaCO_2 44. The disturbance is:

- A. Metabolic alkalosis from loss of HCl
- B. Respiratory alkalosis from hyperventilation
- C. Metabolic acidosis from dehydration
- D. Respiratory acidosis from hypoventilation

[Show answer](#)

A. pH ↑ and HCO_3 ↑ → metabolic alkalosis (ROME: same direction). Cause: loss of gastric HCl. Always pair vomiting with metabolic alkalosis.

Q30. A healthy 20-year-old is having a panic attack, breathing rapidly. ABG: pH 7.52, PaCO_2 26, HCO_3 24. The disturbance is:

- A. Metabolic alkalosis
- B. Respiratory alkalosis
- C. Metabolic acidosis
- D. Respiratory acidosis

[Show answer](#)

B. pH ↑ and PaCO₂ ↓ → respiratory alkalosis (opposite = respiratory). Cause: hyperventilation blows off too much CO₂. Normal HCO₃ confirms acute, uncompensated.

Q31. A patient with severe diarrhea has pH 7.30, HCO₃ 16, PaCO₂ 30. The disturbance and likely cause is:

- A. Respiratory acidosis from hypoventilation
- B. Metabolic acidosis from bicarbonate loss in stool
- C. Metabolic alkalosis from dehydration
- D. Respiratory alkalosis from fever

[Show answer](#)

B. Diarrhea loses HCO₃ → metabolic acidosis. Low PaCO₂ = lungs compensating. Contrast with vomiting (loses HCl → metabolic alkalosis).

Q32. Dropdown. In Kussmaul breathing, the patient breathes [select: fast and deep / slow and shallow / normal rate] because the body is trying to [select: raise CO₂ / lower CO₂ / raise HCO₃], which [select: compensates for metabolic acidosis / causes metabolic alkalosis / has no effect].

[Show answer](#)

Fast and deep · lower CO₂ · compensates for metabolic acidosis. Blowing off CO₂ reduces acid load and pulls pH up toward normal in DKA or severe metabolic acidosis.

Appendix • Top Missteps & Master Mnemonics

Top 15 Week 1 Missteps

#	The trap	What's actually true
1	"Down syndrome is a single-gene mutation."	It's a trisomy (entire extra chromosome 21) from meiotic nondisjunction.
2	"Paternal age drives Down syndrome risk."	Maternal age does — prolonged oocyte meiotic arrest.
3	"Klinefelter is 45,X."	Klinefelter is 47,XXY . Turner is 45,X.
4	"Carrier mothers of Duchenne are affected."	They're usually unaffected (heterozygous X). Affected boys present; brothers/uncles on mom's side may also be affected.
5	"Familial HTN = autosomal dominant."	It's multifactorial (genes + environment). Running in a family doesn't prove Mendelian.
6	"Epigenetic change = DNA mutation."	No — sequence is unchanged; only expression is altered (methylation, histone, miRNA).
7	"Hypertrophy and hyperplasia are the same."	Size vs number. Cardiac muscle cannot hyperplase.
8	"Metaplasia = cancer."	It's adaptive and reversible. Dysplasia is the step toward malignancy.

9	"Apoptosis causes inflammation."	Apoptosis is silent and ATP-requiring . Necrosis causes inflammation.
10	"Hyperkalemia causes U waves."	Hyperkalemia = peaked T waves . U waves = hypokalemia.
11	"Low Ca sedates; high Ca excites."	Opposite. Low Ca = tetany (excitable). High Ca = lethargy and confusion.
12	"Liver failure edema is from high hydrostatic pressure."	It's from low oncotic pressure — the liver can't make albumin.
13	"DKA is respiratory alkalosis because they breathe fast."	DKA is metabolic acidosis ; Kussmaul is compensation, not cause.
14	"Vomiting causes acidosis."	Vomiting loses HCl → metabolic alkalosis . Diarrhea causes acidosis.
15	"Overhydration = dehydration on labs."	Drinking too much plain water causes dilutional hyponatremia , not dehydration.

Master Mnemonic Sheet

Mnemonic	Meaning
ROME	R espiratory O pposite (pH vs CO ₂), M etabolic E qual (pH vs HCO ₃)
AHHAM / AHH-AD	A trophy, H ypertrophy, H yperplasia, A daptive Metaplasia, D ysplasia (abnormal)
"Peaks K"	Peaked T waves = hyperKalemia
"U have low K"	U waves = hypoKalemia
Stones / Bones / Groans / Moans	Hypercalcemia: kidney stones, bone pain, GI pain, psychiatric
"CATs go numb"	Hypo C alcemia causes paresthesia and T etany (Chvostek, Trousseau)
Kussmaul = compensate	Deep, rapid breathing in metabolic acidosis (classic: DKA)
CFTR cascade	Gene → Cl ⁻ channel → water transport → mucus → infection
Hypoxia cascade	↓O ₂ → ↓ATP → pump failure → swelling → lactate → Ca ²⁺ → rupture → necrosis
"Same DNA, different lives"	Identical twin disease discordance = epigenetic

Normal Values Quick Reference

pH 7.35–7.45	PaCO₂ 35–45	HCO₃ 22–26	PaO₂ 80–100
Na 135–145	K 3.5–5.0	Ca 8.5–10.5	Glucose 70–110 fasting
Albumin 3.5–5.0 g/dL	Lactate <2 mmol/L	BUN 7–20 mg/dL	Creatinine 0.6–1.3 mg/dL

Final Study Strategy for Week 1

- **Build the inheritance tree first.** Before any patho, you should be able to look at a pedigree and call AD vs AR vs X-linked in ten seconds. That skill carries the whole genetics section.
- **Learn adaptations in pairs.** Hypertrophy ↔ atrophy (size), hyperplasia ↔ atrophy (number), metaplasia ↔ dysplasia (quality).
- **Memorize two cascades cold.** CFTR → mucus → infection, and hypoxia → ATP → pump fail → necrosis. Ordering questions will draw from these.
- **Use ROME on every ABG.** Don't overthink. pH direction + opposite/same = diagnosis.
- **Learn one classic vignette per disorder.** DKA = young T1, fruity breath, Kussmaul. COPD = older, shallow, confused. Vomiting = alkalosis. Diarrhea = acidosis.
- **Practice ECG electrolyte signs by sight.** Peaked T, flat T with U, wide QRS, prolonged QT (hypoCa).

SCI225 Week 1 Study Guide — built from the Week 1 learning objectives covering Chapters 2, 3, and 4. Made by Dr. Vidya Aggrawal to teach the concepts tested, not the exam questions themselves. Uses the same color-coded format as the midterm review guide. Print for a paper copy, or use in-browser with collapsible practice answers.